Verdix Sustainability Platform — 8-Layer Reference Architecture

Layer 1 · Stakeholder & User Layer

Executives
CEO/CFO · CSO · Board · Investors

Sustainability Analysts
Carbon accountants · ESG reporters · Energy mgrs

Data Stewards & IT
Data owners · MDM · Architects · Security

Compliance & Finance
Regulatory · Audit · Legal · Treasury · FP&A

Procurement & Facilities
Category · Supplier · Contract · Building ops

External Parties
Suppliers (Tier 1–N) · Auditors · Regulators · Customers

Layer 2 · Experience & Presentation

Role-Based Portals

Executive Dashboard | Sustainability Workspace | Supplier Portal | Auditor Portal | Analyst Studio

Interface Channels

Web SPA | Mobile | REST/ GraphQL | Email Notif. | Embedded BI | Convers ational AI

User Experience Services
Personalization · Dashboard Builder · Report Designer · Notifications · Collaboration · Search · Workflow · Offline Sync · WCAG 2.1 · i18n

Layer 3 · Sub-Brand & Application Layer

Verdix Carbon
Scope 1/2/3 · PCF & LCA · Energy mgmt · Emission factors · Carbon ledger · Internal carbon pricing

Verdix ESG
CSRD/ESRS · ISSB S1/S2 · GRI, SASB, CDP, TCFD, TNFD · XBRL tagging · Double materiality · Climate & nature risk

Verdix Compliance
Regulatory tracking · Gap analysis · ICSR · Policy mgmt · SBTi targets · Incident mgmt & CAPA · Data governance

Verdix Supply Chain
Scope 3 · CBAM · EUDR · Supplier portal · N-tier mapping · Risk scoring · Human rights due diligence

Verdix Exchange
Credit registries · Article 6 ITMO · CORSIA EEU · Tokenized trading · Smart-contract retirement · Digital MRV

Verdix Assurance
ISSA 5000 · ISAE 3000 · AA1000AS · ISO 14064-3 · Auditor portal · Evidence mgmt · Immutable audit trail

Verdix Intelligence
MLOps · Predictive analytics · Anomaly detection · GenAI drafting · Agentic AI · Federated learning · BI & KPIs

Shared Application Services
AuthN/AuthZ (SSO, MFA, RBAC/ABAC) · Calculation · Workflow · Notification · Reporting · Search · DMS · Audit log · i18n

Layer 4 · Module & Feature Layer (37 Modules)

Carbon Modules
M01 Carbon Accounting · M04 Energy Mgmt · M08 Product Carbon Footprint · M20 User Experience · M23 Facility & Asset · M33 Mobile & Offline · M37 Financial Integration

ESG Modules
M03 ESG Reporting · M06 Climate & Nature Risk · M25 Waste & Circular Economy · M26 Water & Biodiversity · M28 Stakeholder Engagement

Compliance Modules
M07 Sustainability Governance · M11 Data Governance · M15 Data Lifecycle · M19 RBAC & Security · M22 Regulatory Change · M27 Training · M35 Incident & CAPA

Supply Chain · Exchange · Assurance Modules
M02 Scope 3 & Supply Chain · M24 Supplier Relationship · M05 Carbon Market & Offsets · M31 Blockchain & Tokenization · M21 Audit & Assurance · M34 Document Mgmt

Intelligence — Data Modules
M09 Data Ingestion (500+ connectors) · M10 Data Quality · M12 Master Data Mgmt · M13 Metadata & Ontology · M14 Data Dictionary & Glossary

Intelligence — Analytics & Platform Modules
M16 Analytics & BI · M17 KPI Framework · M18 OKR Mgmt · M29 Workflow Automation · M30 AI/ML & Advanced Analytics · M32 API & Extensibility · M36 Usage Intelligence

Layer 5 · Platform Services

Core Platform Services

Identity & Access
SSO · OAuth/OIDC · MFA · RBAC/ABAC

Workflow & Orchestration
BPMN · Event bus · Saga · SLA

Calculation Engine
Emissions · Energy · PCF/LCA · Factors

Reporting · Search · DMS · i18n
PDF/XLSX/XBRL · Semantic search · Content

AI & Intelligence Services

MLOps Platform
Training · Registry · Feature store · Monitoring

Predictive & NLP
Forecasting · Anomaly · NLQ · Report drafting

Agentic AI Framework
Planning · Tool use · Human-in-the-loop

CV · GNN · Federated · RL
Satellite & methane · Supply-chain graphs

Integration & Interoperability

API Gateway | Integration Platform (500+ connectors) | Event Streaming (Kafka) | Data Exchange (PACT, VC/DID)

Layer 6 · Data Layer

Storage & Management
Operational (PostgreSQL, TimescaleDB, Redis, Neo4j, Vector) · Analytical lakehouse · Document store

MDM · Metadata · Audit
Golden records · Catalog & lineage · Immutable audit trail · Cache, CDN & search index

Data Domains
Emissions · Energy · Supplier · Financial · Asset · Regulatory · Climate · Waste & Circularity · Water & Biodiversity · Social

Layer 7 · Infrastructure & Technology

Cloud Infrastructure

Compute
Kubernetes (EKS/AKS/GKE) · Serverless · Auto-scaling · GPU clusters (AI/ML) · Edge

Storage
Object · Block · File · Archive · Backup & recovery · Replication · Encryption

Networking
VPC/VNet · Load balancers · CDN · DNS · WAF/DDoS · VPN/Private Link · API GW

Container Orchestration
Multi-cloud K8s · Service mesh (Istio/Linkerd) · Helm · Operators · Ingress · Secrets

Security & Compliance

Identity & Access
Zero Trust · SSO/SAML · OAuth 2.0/OIDC · MFA · RBAC/ABAC · PAM · Secrets · Certs

Data Protection
AES-256 · TLS 1.3 · KMS/HSM · Masking · Tokenization · Post-quantum · Homomorphic · DLP

Network Security
Firewalls · IDS/IPS · WAF · DDoS · API security · Micro-segmentation · ZTNA

Compliance & Governance
SOC 1/2 Type 2 · ISO 27001/27017/27018 · GDPR · CCPA · SOX · HIPAA · ISSA 5000 · ISAE 3000

DevOps & Operations

CI/CD Pipeline
GitOps · Automated testing · Blue-green · Canary · Feature flags · IaC (Terraform/Pulumi) · ArgoCD

Observability
Logging (ELK/OpenSearch) · Metrics (Prometheus/Grafana) · Tracing · Alerting · SLI/SLO

AIOps
Anomaly detection · Predictive maintenance · Auto-remediation · Capacity & cost · Self-healing

Disaster Recovery
Multi-region · Backup & restore · RTO/RPO · Chaos engineering · Failover tests · Geo redundancy

Layer 8 · External Integration

Enterprise Systems

ERP
SAP · Oracle · Dynamics · NetSuite

CRM
Salesforce · HubSpot · Zoho

HRIS
Workday · SuccessFactors · Oracle HCM

Procurement
SAP Ariba · Coupa · Jaggaer · GEP

Finance
SAP S/4 · Oracle Fusion · Workday Adaptive

PLM
Siemens Teamcenter · Dassault · PTC · Autodesk

Data Providers & Registries

Emission Factors
DEFRA · EPA · IEA · IPCC · Climatiq · Ecoinvent

Utility Data
Utility portals · Urjanet · Arcadia · Energy Star · NREL · EIA

Travel Data
Amex GBT · SAP Concur · TravelPerk · TripActions

Carbon Registries
Verra · Gold Standard · ACR · Puro.earth · ICVCM · CORSIA

Rating Agencies
MSCI · Sustainalytics · S&P Global · CDP · EcoVadis · Refinitiv

Satellite Imagery
Planet · Maxar · Airbus · ICEYE · GHGSat

IoT & Sensor Networks

Energy Meters
Smart & sub-meters · Power quality · Renewable generation

Emissions Sensors
CEMS · Methane · CO2 · VOC · Flare monitoring

Water Meters
Flow · Quality · Pressure · Level sensors

Waste Sensors
Fill level · Weight · Compactors · Cameras

Building BMS
HVAC · Lighting · Occupancy · Temp/humidity · Air quality

Vehicle Telematics
GPS · Fuel consumption · Engine data · Mileage · Route

External Stakeholder Systems

Supplier Systems
Supplier portals · EDI · API · Email · Mobile app · Web forms

Auditor Systems
Audit mgmt · Evidence mgmt · Workpapers · Certification

Regulator Portals
CDP · SEC · EU business registers · National registries · Exchanges

Investor Portals
ESG ratings · Bloomberg · MSCI · Sustainalytics · Refinitiv · CDP · TCFD

Customer Portals
Product carbon labels · Customer RFPs · Supply-chain surveys

Industry Associations
Benchmarking · Best-practice sharing